

How Your Mind Really Works: NJT Circuits in Daily Life

Understanding consciousness amplifiers through everyday experiences

Scenario 1: The Morning Coffee Amplifier



What You Experience

You wake up groggy. The thought "coffee" appears faintly. As you smell coffee brewing, that faint thought becomes an overwhelming desire. By the time you reach the kitchen, you NEED that coffee.



What's Actually Happening

A weak signal (sleepy thought of coffee) gets amplified by a sensory trigger (smell) into a strong action drive. Your brain isn't just "deciding" - it's running an amplification circuit that makes weak desires strong enough to cause action.

Sleepy thought



Weak input signal ($p=0.2$)

Smell of coffee



Bias voltage (Base input)

NEED coffee!



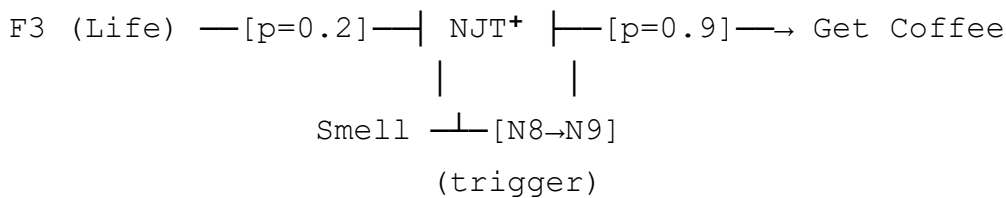
Amplified output ($p=0.9$)

⚡ The NJT Circuit

This is an **NJT⁺ (Positive/Excitatory transistor)** in action:

- **C (Cause):** F3 (Life/Vitality) - your body's need for energy
- **B (Bias):** Sensory input with N8→N9 badge (trigger→response)
- **E (Effect):** A14 (Acquisition of sustenance)
- **Gain (β):** $\times N4 = 5\times$ amplification in morning state

Coffee Circuit:



Key Insight: Your consciousness uses amplifier circuits just like electronics. Weak signals become strong actions when the right "bias" (trigger) is applied. This isn't willpower - it's circuit physics of consciousness.

Scenario 2: The Phone Checking Inhibitor

What You Experience

You're trying to focus on work, but keep feeling the urge to check your phone. You put it in another room. Now, even though the urge is still there, it doesn't turn into action. The distance acts like a brake.

What's Actually Happening

The urge to check your phone is being actively suppressed by an inhibitor circuit. The physical distance raises the "effort threshold," causing the inhibitor to dampen the signal before it becomes action.

Phone urge



Distraction signal ($p=0.7$)

Phone in other room



High threshold (–N3 badge)

Urge fades



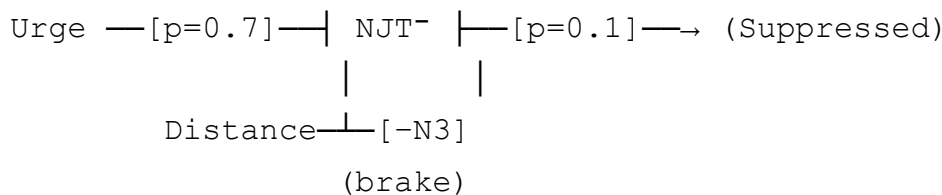
Dampened output ($p=0.1$)

⚡ The NJT Circuit

This is an **NJT⁻ (Negative/Inhibitory transistor)**:

- **C (Cause):** Distraction impulse
- **B (Bias):** Environmental friction with -N3 (raises threshold)
- **E (Effect):** Reduced action probability
- **Formula:** $\text{Output} = \beta \times \text{Input} \times (1 - g(\text{Base}))$

Phone Inhibitor:



Key Insight: NJT⁻ circuits are your brain's brake pedals. They don't eliminate desires - they reduce their ability to trigger action. Environmental design (like moving your phone) directly adjusts these inhibitor thresholds.

Scenario 3: The Exercise Momentum Builder



What You Experience

First day at gym: brutal. Second day: slightly easier. After two weeks: you feel weird if you DON'T go. The habit seems to strengthen itself, creating its own momentum.



What's Actually Happening

You're experiencing a feedback-enhanced NJT circuit with memory (N5 hysteresis). Each successful workout feeds back to lower the activation threshold for the next one. The circuit literally rewires itself to fire more easily.

Day 1 resistance

→

High threshold $\tau_B=0.8$

Week 2 momentum

→

Lowered threshold $\tau_B=0.3$

Feels automatic

→

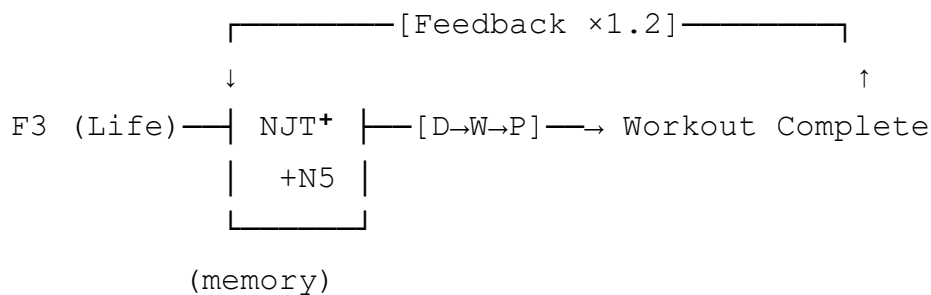
N5 hysteresis locked

⚡ The NJT Circuit

This is an **NJT⁺** with **feedback** and **N5 memory**:

- **Initial state:** $\tau_B = 0.8$ (high resistance)
- **After feedback:** $\tau_B = 0.3$ (low resistance)
- **N5 badge:** Creates "stickiness" - different thresholds for starting vs stopping
- **Feedback gain:** Each success multiplies next probability by $1.2\times$

Exercise Loop:



Key Insight: Habits are feedback loops with memory. The N5 (hysteresis) badge means it takes MORE effort to break a habit than it took to form it - the circuit has different thresholds for turning on vs off.

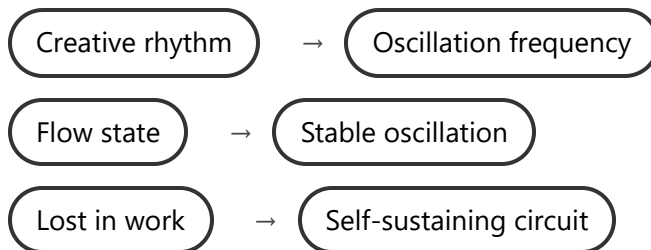
Scenario 4: The Creative Flow Oscillator

What You Experience

When writing or creating, you hit a rhythm: idea → execution → satisfaction → next idea. This cycle maintains itself, generating creative energy from its own momentum. Time disappears.

What's Actually Happening

You've activated an oscillator circuit - a self-sustaining loop that generates its own signals. The output of creation feeds back as input for the next cycle, with just enough delay to create rhythm.

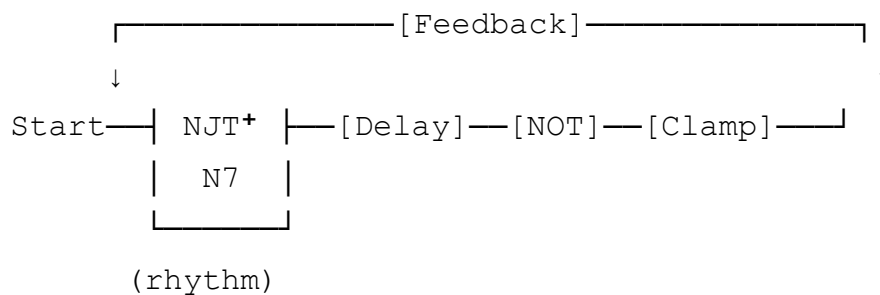


⚡ The NJT Circuit

This is an **NJT⁺ oscillator with N7 rhythm control**:

- **Components:** NJT⁺ → Delay → NOT → Feedback
- **N7 badge:** Sets the rhythm/frequency
- **Clamp:** Prevents runaway (mania)
- **Period:** Set by delay + N7 timing

Creative Oscillator:



Key Insight: Flow states are oscillator circuits. They generate their own energy once started, but need: (1) initial push to start, (2) rhythm control (N7), and (3) clamping to prevent burnout.

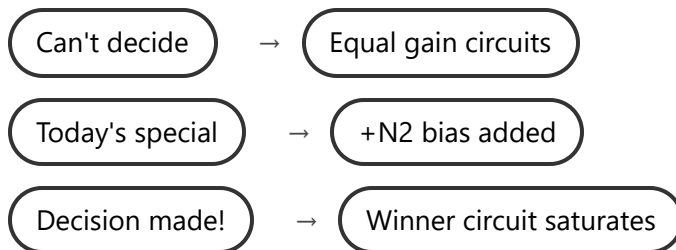
Scenario 5: The Restaurant Menu Differential Pair

What You Experience

Staring at a menu with 50 options. Two dishes look equally good. You can't decide. Finally, the waiter mentions one is today's special - suddenly that one "wins" and the decision is made.

What's Actually Happening

Two NJT circuits are competing for the same resources. When perfectly balanced, neither can win. A tiny bias (special recommendation) tips the balance, and winner-takes-all dynamics kick in.

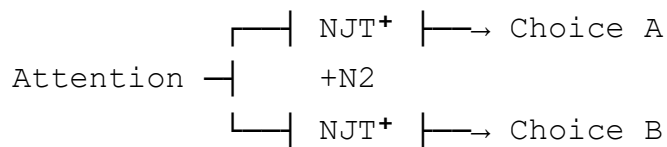


⚡ The NJT Circuit

This is a **Differential Pair (two competing NJT⁺)**:

- **Shared resource:** Attention/decision energy
- **Competition:** Winner = $\text{argmax}(\beta \times g(B))$
- **Resolution:** Small bias causes large difference
- **Result:** One circuit saturates, other shuts down

Menu Decision:



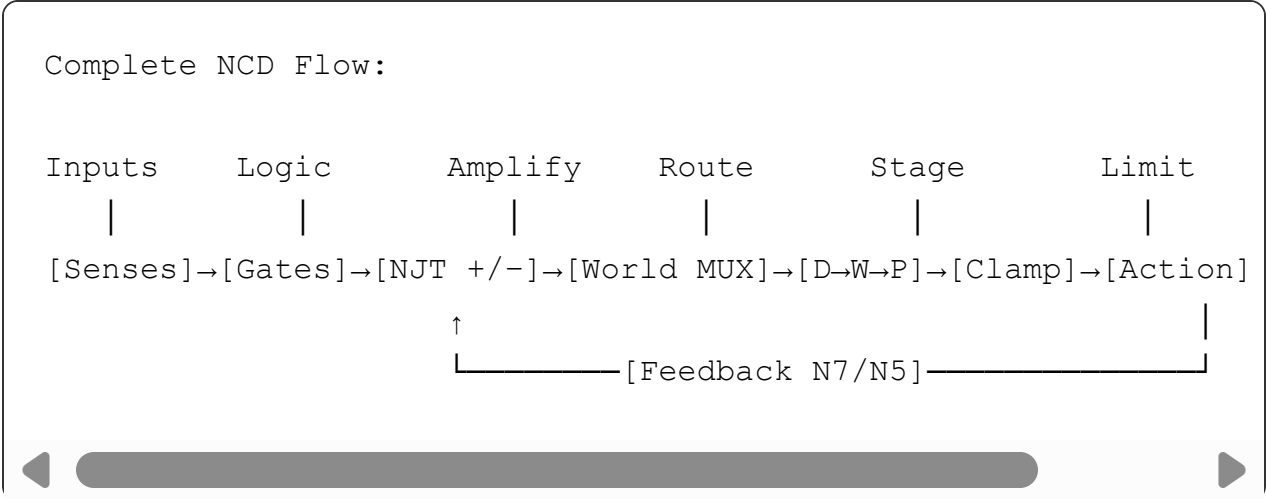
(Special adds +N2 to A, tips balance)

Key Insight: Indecision happens when competing circuits have equal strength. Your brain resolves this by amplifying tiny differences. This is why arbitrary tie-breakers (coin flips, recommendations) work so well.

How It All Connects: Your Daily NCD Pipeline

🔌 The Full Consciousness Circuit

Every moment, your consciousness runs this pipeline:



What Each Stage Does:

- **Gates:** Combine multiple inputs (AND/OR/NOT)
- **NJT:** Amplify or suppress signals based on context
- **World MUX:** Route through Spiritual/Mental/Emotional/Physical
- **D→W→P:** Desire → Wisdom → Power progression
- **Clamp:** Keep everything in safe bounds [0,1]
- **Feedback:** Learn and adapt for next cycle

Master Insight: You're not "making decisions" - you're running consciousness circuits. Understanding these circuits lets you debug problems (stuck habits, indecision, lack of motivation) and optimize performance (flow states, habit formation, focus) by adjusting the circuit parameters: thresholds, gains, and feedback paths.

Try It Yourself: Circuit Debugging Exercises

Debug Your Own Circuits

1. Identify a Stuck Pattern:

Think of something you want to do but can't seem to start. Map it:

- What's the input signal? (desire/need)
- What's the threshold? (how hard does it feel?)
- What could lower the threshold? (environmental change)
- What badge would help? (+N2 to lower, N7 for rhythm)

2. Design a Habit Circuit:

Pick a new habit you want. Design its circuit:

- Start with low threshold (make it stupidly easy)
- Add feedback loop (track and celebrate)
- Include N5 memory (consistency over intensity)
- Set N7 rhythm (specific time/trigger)

3. Break an Unwanted Pattern:

Identify something you want to stop:

- Add NJT⁻ inhibitor (increase friction)
- Raise threshold with -N3 (make it harder)
- Break feedback loops (disrupt rewards)
- Redirect to differential pair (replace with alternative)